MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RECEIVER, TRANSLATOR, DIGITAL, KEY CODE DLC

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic translator receiver module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.168 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be DLC.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/313-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc and (V_{DD}) of -5.0 ± 0.25 V dc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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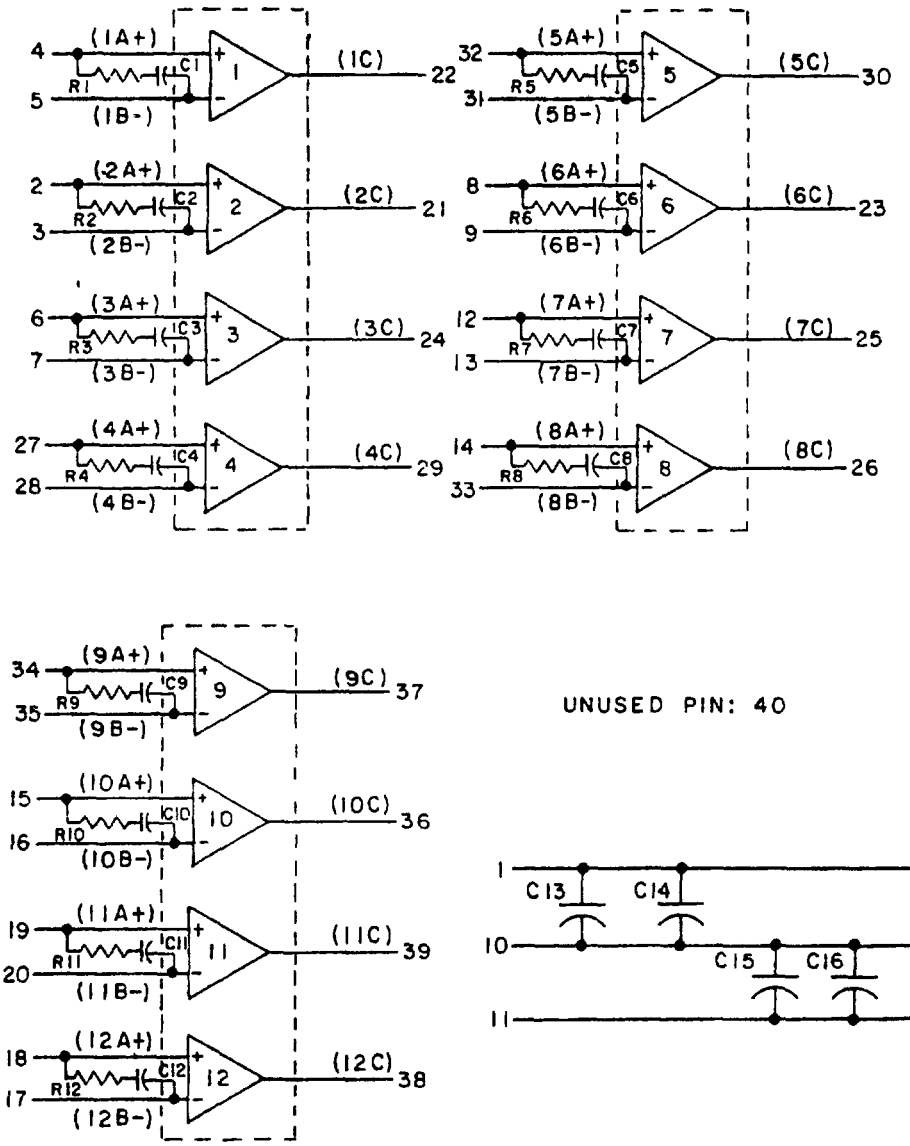


FIGURE 1 DLC module function diagram, translator receiver.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1	Isolation	III	S	10	∞	1	∞	10	∞	1	∞	MΩ
A2	Isolation, unused pin	III	Q	10	∞	1	∞	10	∞	1	∞	MΩ
B1,B2	Power sequencing	IV	S	◄----------Module must pass all remaining tests----------►								
C1-C12	Input capacitance check	V	Q	0.0074	0.0094	0.0074	0.0094	0.0074	0.0094	0.0074	0.0094	µF
D1-D12	Output short circuit current	VI	X	10	110	10	110	10	110	10	110	mA
E1-E12	Output voltage, low level	VII	X	-500	400	-500	400	-500	400	-500	400	mV
E13-E24	Output voltage, high level	VII	X	2.4	5.0	2.4	5.0	2.4	5.0	2.4	5.0	V
F1-F36	Common mode rejection, low level	VIII	S	-500	400	-500	400	-500	400	-500	400	mV
G1-G24	Common mode rejection, high level	IX	S	2.4	5.0	2.4	5.0	2.4	5.0	2.4	5.0	V
H1-H12	Input current, low level	X	X	-100	-500	-100	-500	-100	-500	-100	-500	µA
J1-J12	Input current, high level	XI	X	0.1	1.5	0.1	1.5	0.1	1.5	0.1	1.5	mA
K1	Power supply current, low level, V_{CC}	XII	S	50	125	50	125	50	125	50	125	mA
K2	Power supply current, low level, V_{DD}	XII	S	70	165	70	165	70	165	70	165	mA
K3	Power supply current, high level, V_{CC}	XII	S	50	125	50	125	50	125	50	125	mA
K4	Power supply current, high level, V_{DD}	XII	S	70	165	70	165	70	165	70	165	mA
L1-L12	Transition time (t_{TLH})	XIII	S	10	125	10	125	10	125	10	125	ns
M1-M12	Transition time (t_{THL})	XIV	S	10	125	10	125	10	125	10	125	ns

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See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N1-N12	Propagation delay time (t_{PLH})	XV	S	10	200	10	200	10	200	10	200	ns
P1-P12	Propagation delay time (t_{PHL})	XVI	S	10	200	10	200	10	200	10	200	ns
Q1,Q2	Power supply transient amplitude	XVII	Q	0	400	0	400	0	400	0	400	mV
R1-R12	Output crosstalk, low level	XVIII	Q	0	400	0	400	0	400	0	400	mV
S1-S12	Output crosstalk, high level	XVIII	Q	0	400	0	400	0	400	0	400	mV
T1	Power supply transient rise time, V_{CC}	XIX	S	0.5	1.5	0.5	1.5	0.5	1.5	0.5	1.5	ms
T2	Power supply transient rise time, V_{DD}	XIX	S	0.5	1.5	0.5	1.5	0.5	1.5	0.5	1.5	ms

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. If test condition A is part of a sequence of tests, the order of testing shall be as follows: A where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through T shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test conditions T1 and T2, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through T shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through T shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

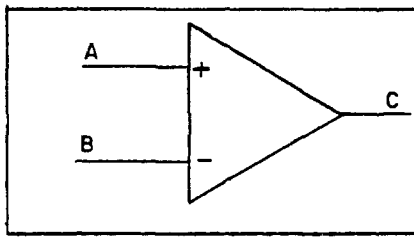
6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Module function. The translator receiver module, contains twelve interface line receiver circuits. It is designed to translate logic signals from negative voltage levels to positive voltage levels. The DLC module is intended to be used with translator-driver module DLB (MIL-M-28787/312), or equivalent. The output will be a low level when both inputs are open. The input circuit presents a terminal impedance to the line of approximately 160Ω in series with 0.008 µF. When used with twisted shielded pair, the high side of the line should be connected to the plus (+) input, the slow side of the line should be connected to the minus (-) input, and the shield should be connected to the chassis.

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6.3.1 Input to output relationship. Each receiver shall have the input to output relationship of table II for positive logic:

TABLE II. Input to output relationship for positive logic.



INPUT	OUTPUT
$V_A - V_B$	C
≥ (−) 1.9 V	0
≤ (−) 1.1 V	1

6.3.2 Parallel outputs. Paralleling of outputs is not permissible.

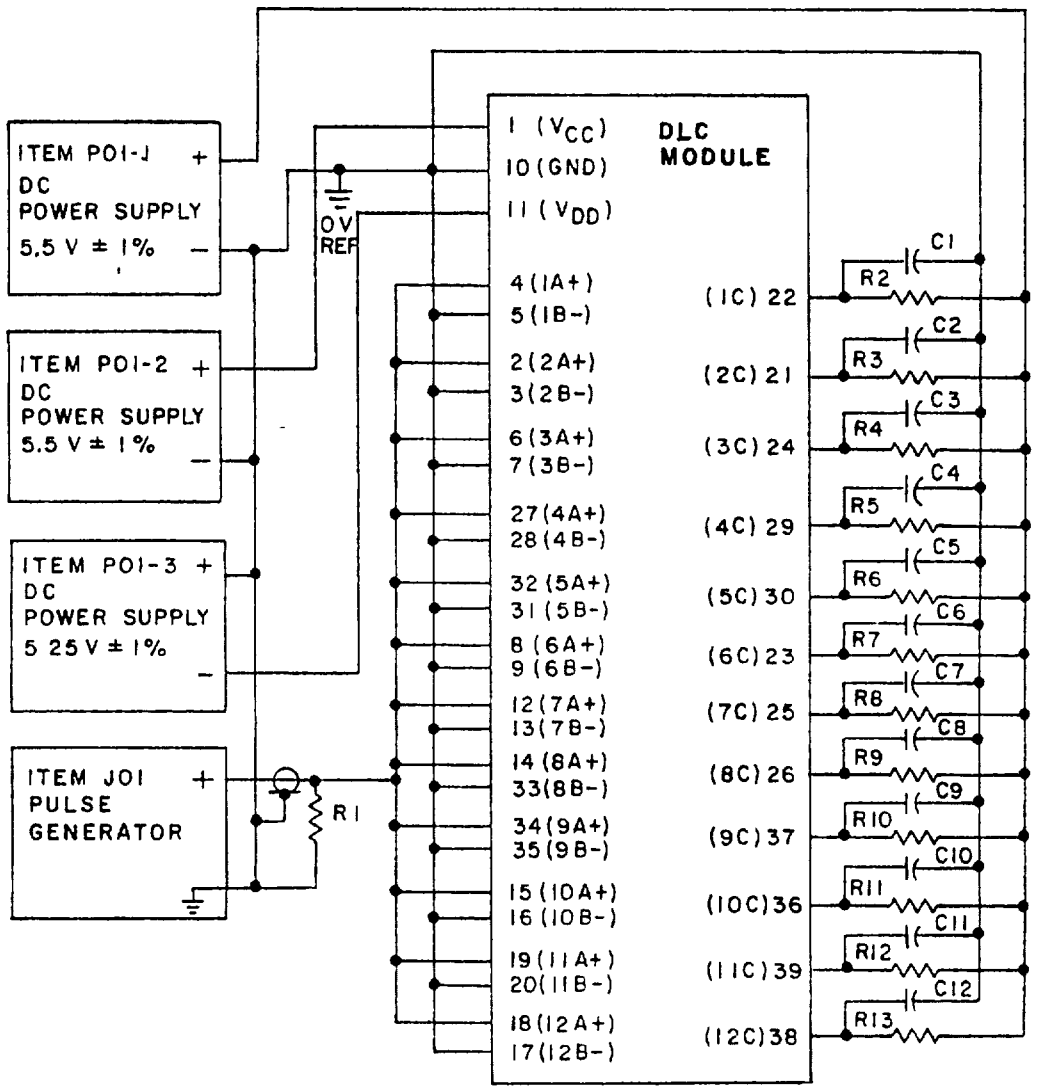
6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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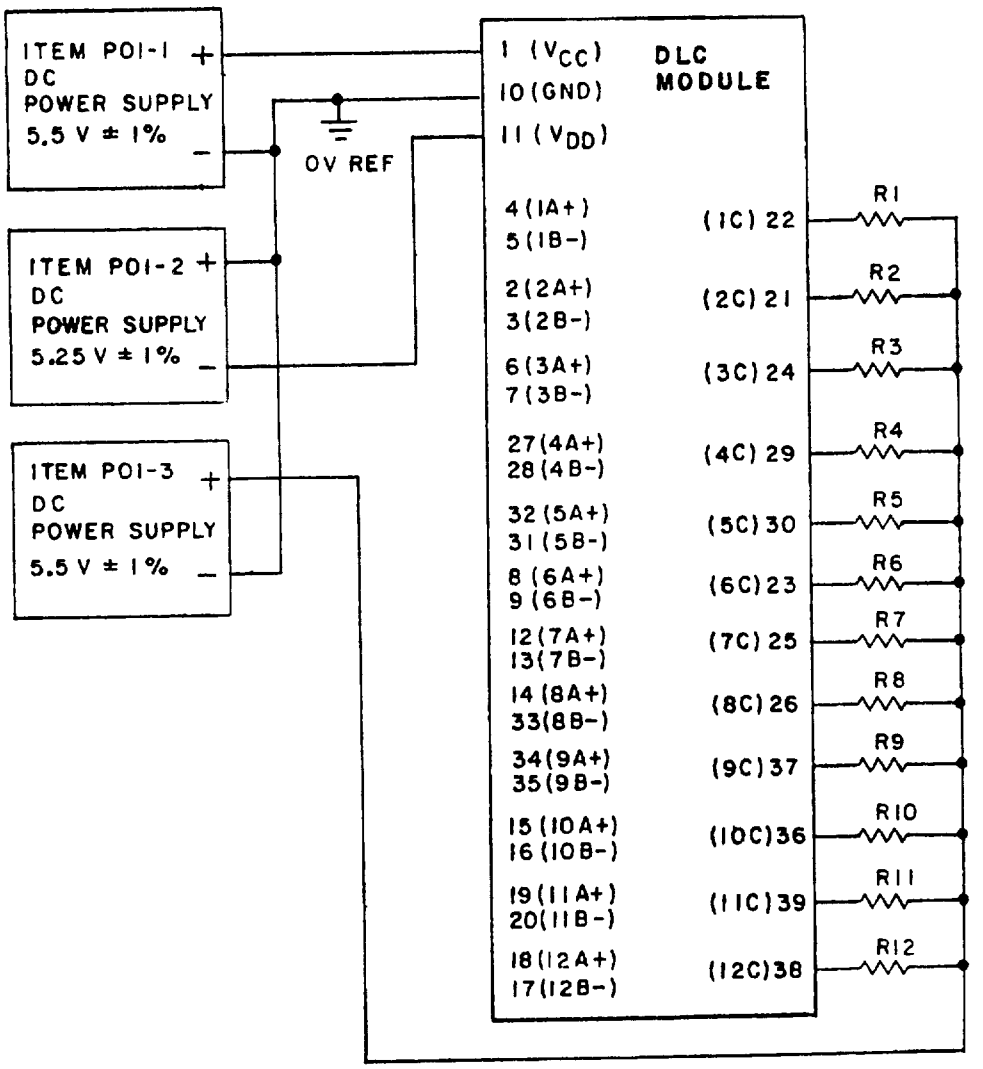
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2-R13 = 3.16 kΩ ±1%, 1/4 W.
3. C1-C12 = 50 pF ±20%.
4. Item J01 setup:
   PA = -3 V ±10%.
   Low level = 0.0 ±0.2 V.
   High level = -3 V ±10%.
   PW = 1 µs ±10%/
   PRR = 500 kHz ±10%.
   $t_{TLH} = t_{THL} = < 100$ ns.

FIGURE 2. Operating temperature and life test circuit

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NOTE: R1-R12 = 3.16 kΩ ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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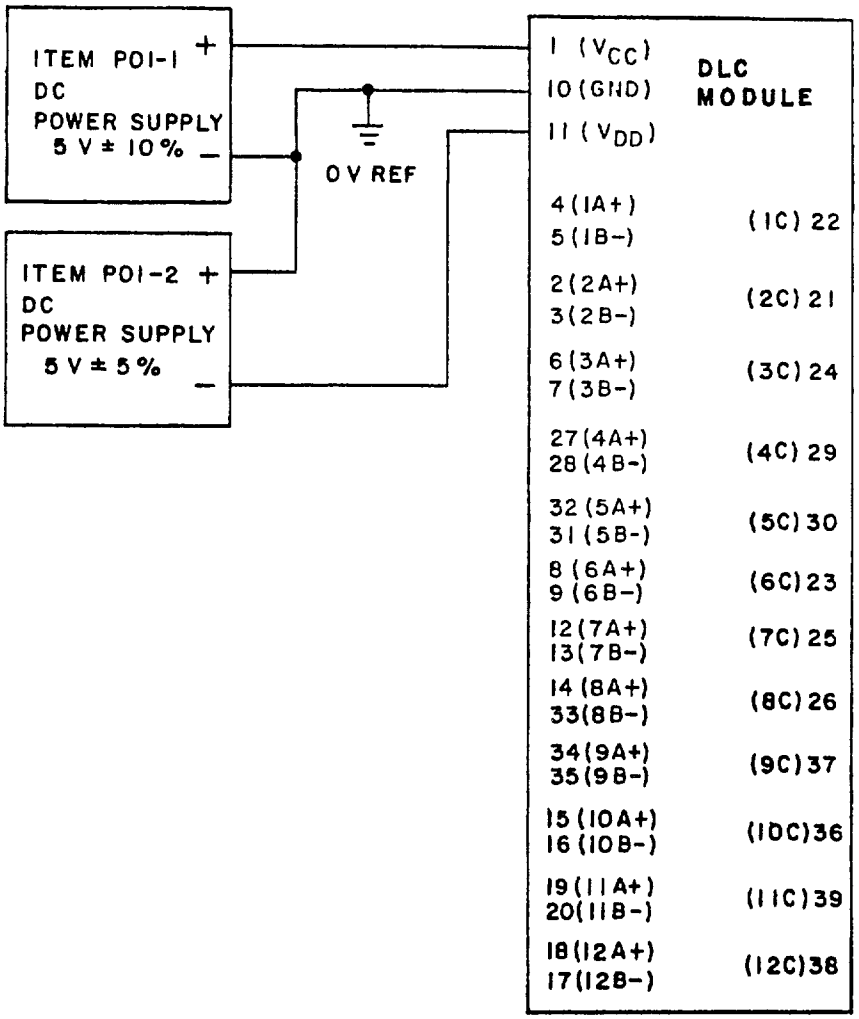


FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

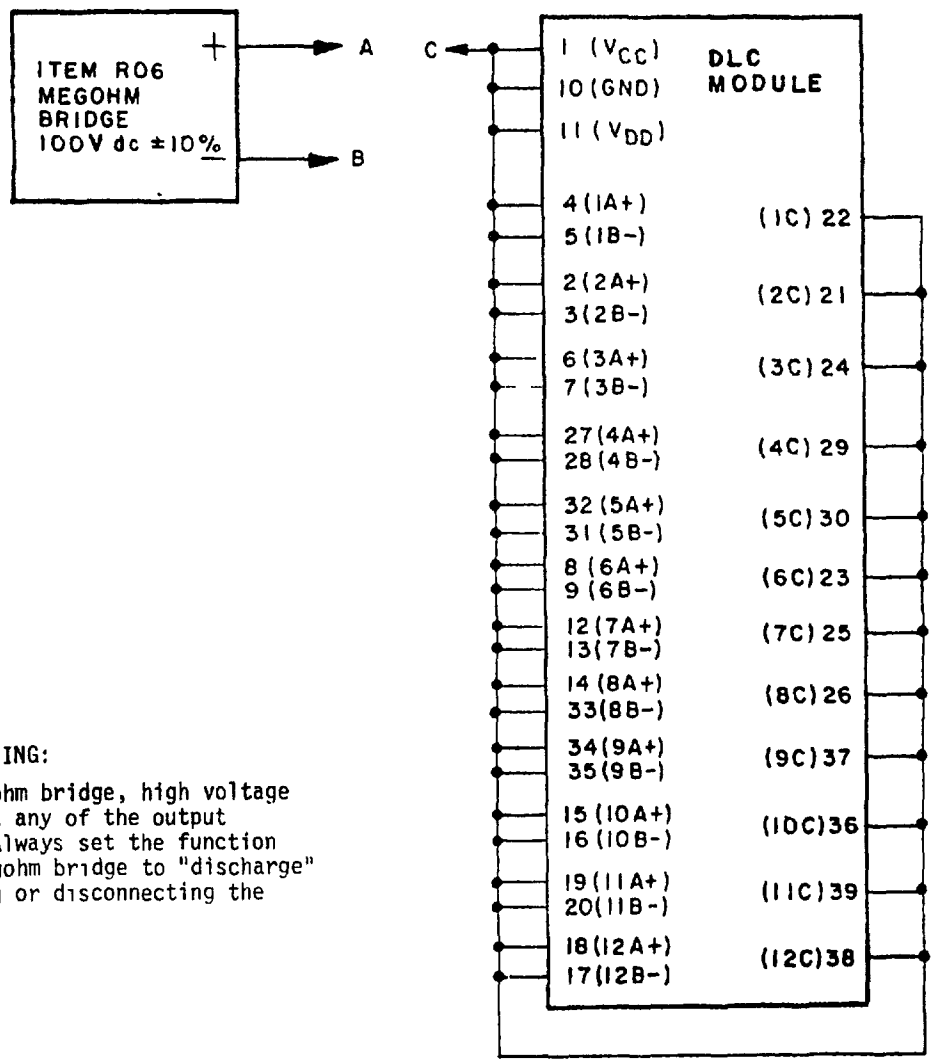
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C, 40	Frame	10	∞	MΩ	5
A2	Isolation, unused pin	C	40	10	∞	MΩ	5

TABLE IV. Power sequencing test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Power sequencing	1	Open	1/	1/	2/	6
B2	Power sequencing	Open	11	1/	1/	2/	6

1/ The module shall pass all remaining tests.
2/ The test connections shall be made for a maximum of 60 seconds.

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads

NOTE: For test condition A1 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table III.

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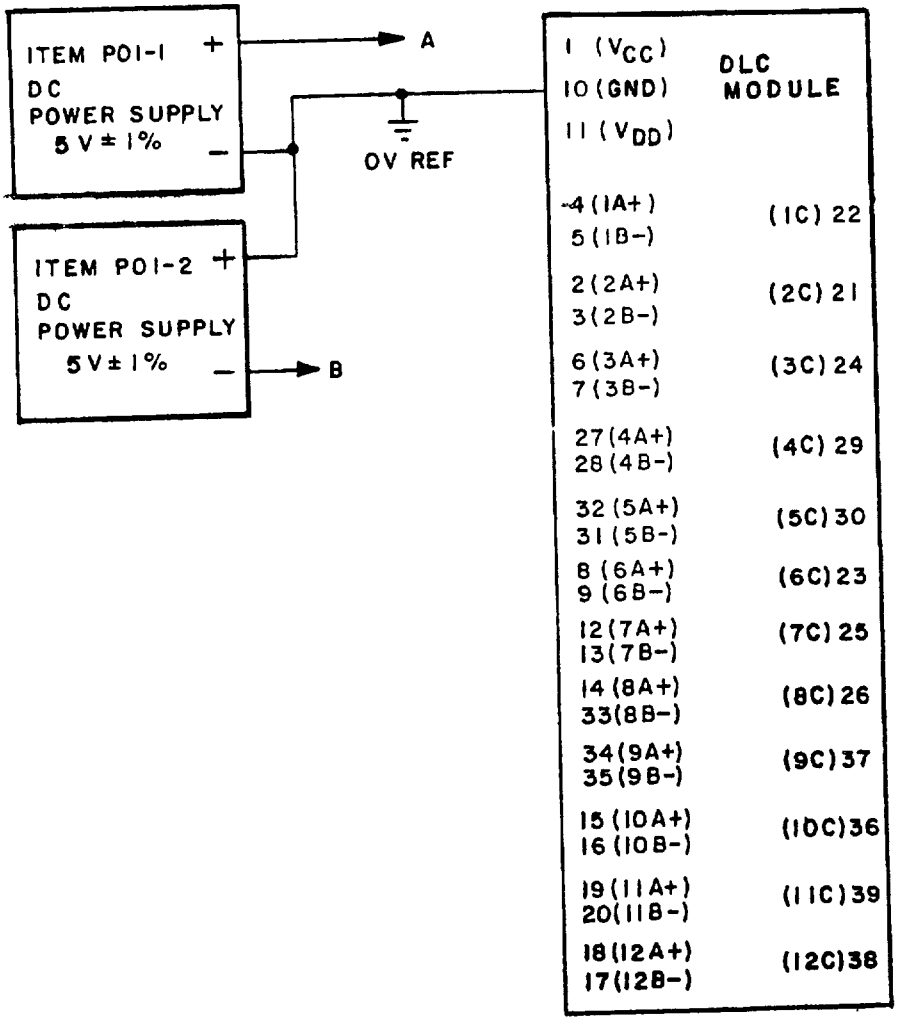


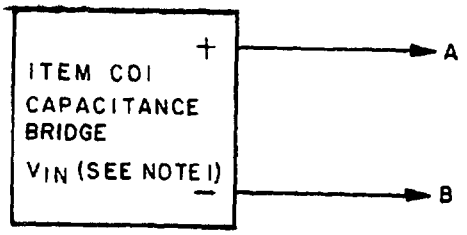
FIGURE 6. Power sequencing test circuit for table IV.

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* TABLE V. Input capacitance check test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	1A,1B	Input capacitance check	4	5	0.0074	0.0094	µF	7
C2	2A,2B	"	2	3	"	"	"	"
C3	3A,3B	"	6	7	"	"	"	"
C4	4A,4B	"	27	28	"	"	"	"
C5	5A,5B	"	32	31	"	"	"	"
C6	6A,6B	"	8	9	"	"	"	"
C7	7A,7B	"	12	13	"	"	"	"
C8	8A,8B	"	14	33	"	"	"	"
C9	9A,9B	"	34	35	"	"	"	"
C10	10A,10B	"	15	16	"	"	"	"
C11	11A,11B	"	19	20	"	"	"	"
C12	12A,12B	"	18	17	"	"	"	"

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	DLC MODULE
1 (V_{CC}) 10 (GND) 11 (V_{DD})	
4 (1A+) 5 (1B−)	(1C) 22
2 (2A+) 3 (2B−)	(2C) 21
6 (3A+) 7 (3B−)	(3C) 24
27 (4A+) 28 (4B−)	(4C) 29
32 (5A+) 31 (5B−)	(5C) 30
8 (6A+) 9 (6B−)	(6C) 23
12 (7A+) 13 (7B−)	(7C) 25
14 (8A+) 33 (8B−)	(8C) 26
34 (9A+) 35 (9B−)	(9C) 37
15 (10A+) 16 (10B−)	(10C) 36
19 (11A+) 20 (11B−)	(11C) 39
18 (12A+) 17 (12B−)	(12C) 38

NOTES:
1. Set V_{IN} to 0.2 V at "unknown" terminals at null.
2. Measure the values of Q and G at 1 kHz.
3. Use formula $C_{IN} = \frac{QG}{2\pi F}$ to calculate value of input capacitance.

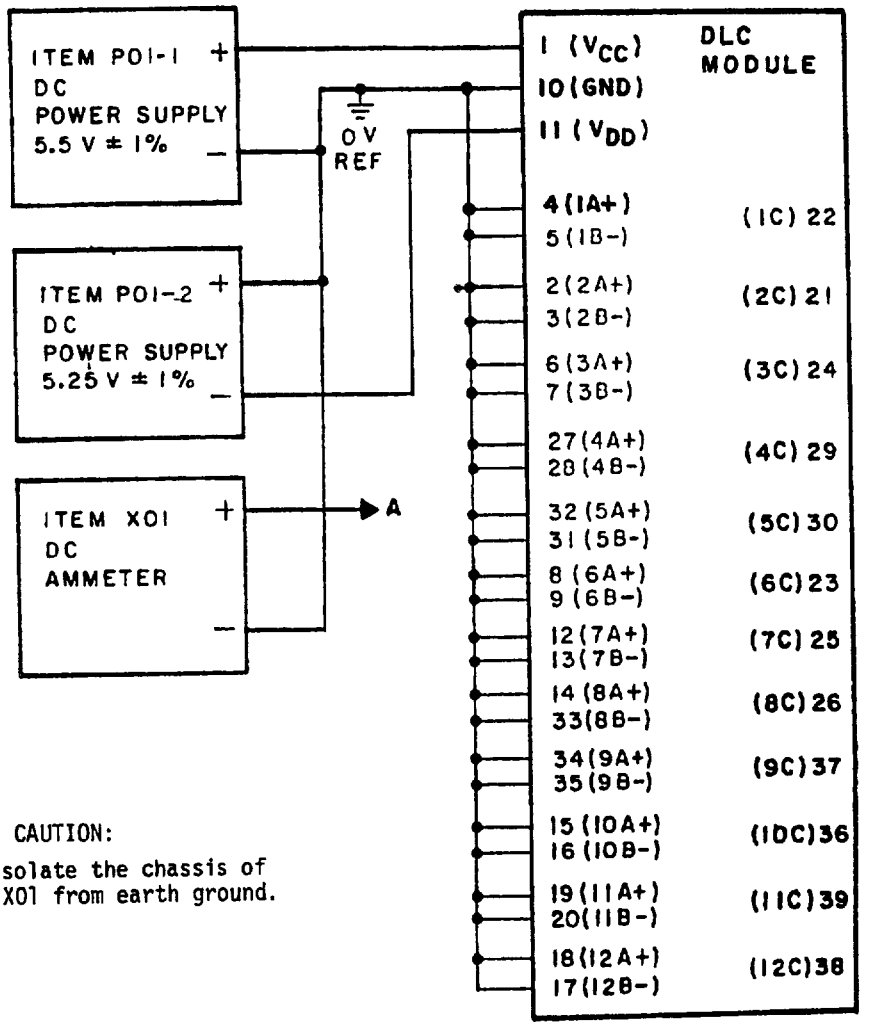
FIGURE 7. Input capacitance check test circuit for table V.

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* TABLE VI. Output short circuit current test.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
D1	1C	Output short circuit current	22	10	110	mA	8
D2	2C	"	21	"	"	"	"
D3	3C	"	24	"	"	"	"
D4	4C	"	29	"	"	"	"
D5	5C	"	30	"	"	"	"
D6	6C	"	23	"	"	"	"
D7	7C	"	25	"	"	"	"
D8	8C	"	26	"	"	"	"
D9	9C	"	37	"	"	"	"
D10	10C	"	36	"	"	"	"
D11	11C	"	39	"	"	"	"
D12	12C	"	38	"	"	"	"

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE: Only one output shall be shorted at a time. The duration shall be 1 second maximum.

* FIGURE 8 Output short circuit current test circuit for table VI.

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* TABLE VII. Output voltage (low-level and high-level) test.

Test condition	Parameter	Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
E1	1C Output voltage, low level	B,5	A	4	Open	Open	22	-500	400	mV	9
E2	2C "	B,3	"	2	"	"	21	"	"	"	"
E3	3C "	B,7	"	6	"	"	24	"	"	"	"
E4	4C "	B,28	"	27	"	"	29	"	"	"	"
E5	5C "	B,31	"	32	"	"	30	"	"	"	"
E6	6C "	B,9	"	8	"	"	23	"	"	"	"
E7	7C "	B,13	"	12	"	"	25	"	"	"	"
E8	8C "	B,33	"	14	"	"	26	"	"	"	"
E9	9C "	B,35	"	34	"	"	37	"	"	"	"
E10	10C "	B,16	"	15	"	"	36	"	"	"	"
E11	11C "	B,20	"	19	"	"	39	"	"	"	"
E12	12C "	B,17	"	18	"	"	38	"	"	"	"
E13	1C Output voltage, high level	B,5	"	4	"	"	22	2.4	5.0	V	"
E14	2C "	B,3	"	2	"	"	21	"	"	"	"
E15	3C "	B,7	"	6	"	"	24	"	"	"	"
E16	4C "	B,28	"	27	"	"	29	"	"	"	"
E17	5C "	B,31	"	32	"	"	30	"	"	"	"
E18	6C "	B,9	"	8	"	"	23	"	"	"	"
E19	7C "	B,13	"	12	"	"	25	"	"	"	"

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* TABLE VII. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
E20	8C Output voltage, high level	B,33	A	14	Open	Open	26	2.4	5.0	V	9
E21	9C "	B,35	"	34	"	"	37	"	"	"	"
E22	10C "	B,16	"	15	"	"	36	"	"	"	"
E23	11C "	B,20	"	19	"	"	39	"	"	"	"
E24	12C "	B,17	"	18	"	"	38	"	"	"	"

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TABLE VIII. Common mode rejection (low-level) test.

Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
F1	1A,1B	Common mode rejection, low level	C,E	4	A	5	A	22	-500	400	mV	9
F2	2A,2B	"	"	2	"	3	"	21	"	"	"	"
F3	3A,3B	"	"	6	"	7	"	24	"	"	"	"
F4	4A,4B	"	"	27	"	28	"	29	"	"	"	"
F5	5A,5B	"	"	32	"	31	"	30	"	"	"	"
F6	6A,6B	"	"	8	"	9	"	23	"	"	"	"
F7	7A,7B	"	"	12	"	13	"	25	"	"	"	"
F8	8A,8B	"	"	14	"	33	"	26	"	"	"	"
F9	9A,9B	"	"	34	"	35	"	37	"	"	"	"
F10	10A,10B	"	"	15	"	16	"	36	"	"	"	"
F11	11A,11B	"	"	19	"	20	"	39	"	"	"	"
F12	12A,12B	"	"	18	"	17	"	38	"	"	"	"
F13	1A,1B	"	B,D	A	4	A	5	22	"	"	"	"
F14	2A,2B	"	"	"	2	"	3	21	"	"	"	"
F15	3A,3B	"	"	"	6	"	7	24	"	"	"	"
F16	4A,4B	"	"	"	27	"	28	29	"	"	"	"
F17	5A,5B	"	"	"	32	"	31	30	"	"	"	"
F18	6A,6B	"	"	"	8	"	9	23	"	"	"	"
F19	7A,7B	"	"	"	12	"	13	25	"	"	"	"

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TABLE VIII. Common mode rejection (low-level) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
F20	8A,8B	Common mode rejection, low level	B,D	A	14	A	33	26	-500	400	mV	9
F21	9A,9B	"	"	"	34	"	35	37	"	"	"	"
F22	10A,10B	"	"	"	15	"	16	36	"	"	"	"
F23	11A,11B	"	"	"	19	"	20	39	"	"	"	"
F24	12A,12B	"	"	"	18	"	17	38	"	"	"	"
F25	1A,1B	"	Open	Open	Open	Open	Open	22	"	"	"	"
F26	2A,2B	"	"	"	"	"	"	21	"	"	"	"
F27	3A,3B	"	"	"	"	"	"	24	"	"	"	"
F28	4A,4B	"	"	"	"	"	"	29	"	"	"	"
F29	5A,5B	"	"	"	"	"	"	30	"	"	"	"
F30	6A,6B	"	"	"	"	"	"	23	"	"	"	"
F31	7A,7B	"	"	"	"	"	"	25	"	"	"	"
F32	8A,8B	"	"	"	"	"	"	26	"	"	"	"
F33	9A,9B	"	"	"	"	"	"	37	"	"	"	"
F34	10A,10B	"	"	"	"	"	"	36	"	"	"	"
F35	11A,11B	"	"	"	"	"	"	39	"	"	"	"
F36	12A,12B	"	"	"	"	"	"	38	"	"	"	"

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TABLE IX. Common mode rejection (high-level) test.

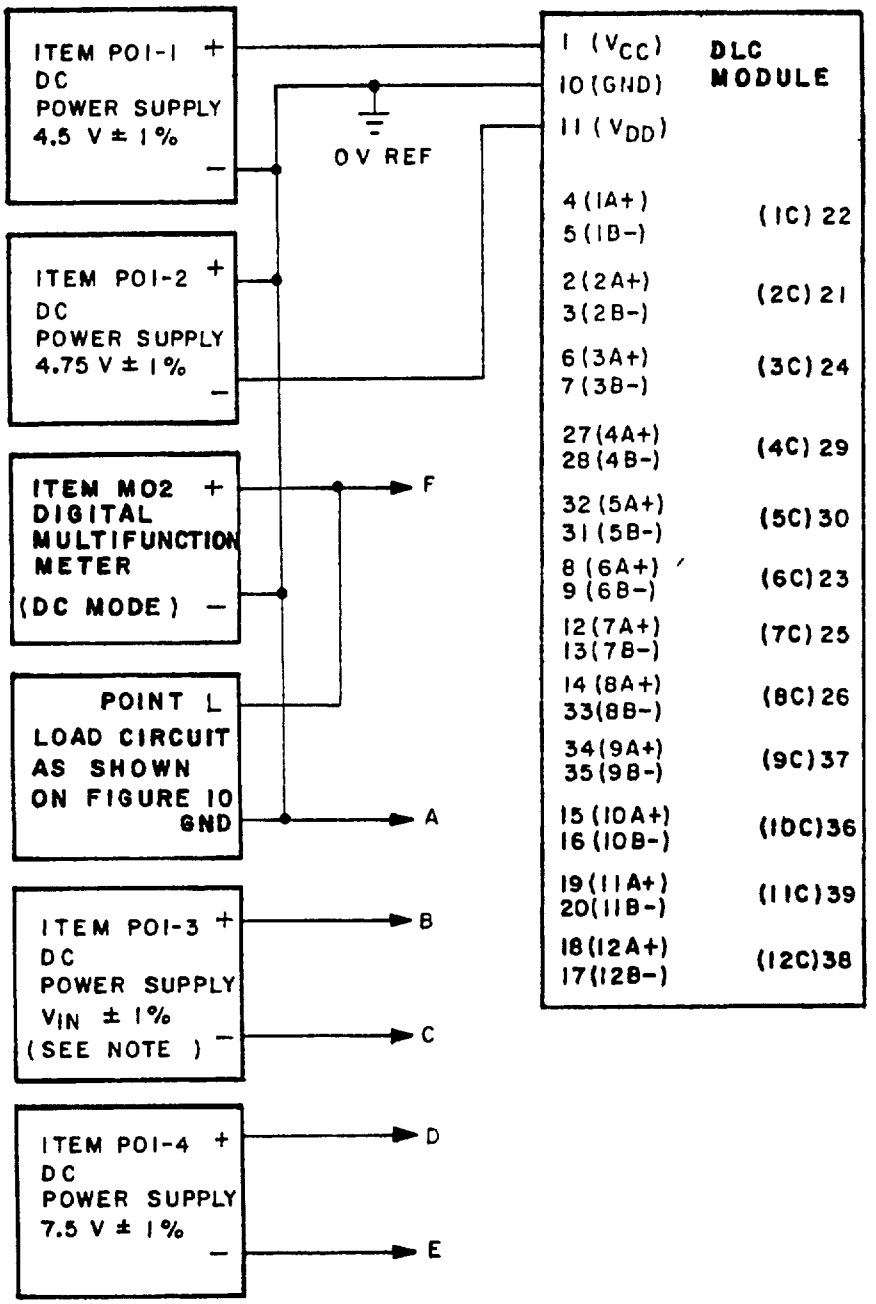
Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
G1	1A,1B	Common mode rejection, high level	C,E	4	A	5	A	22	2.4	5.0	V	9
G2	2A,2B	"	"	2	"	3	"	21	"	"	"	"
G3	3A,3B	"	"	6	"	7	"	24	"	"	"	"
G4	4A,4B	"	"	27	"	28	"	29	"	"	"	"
G5	5A,5B	"	"	32	"	31	"	30	"	"	"	"
G6	6A,6B	"	"	8	"	9	"	23	"	"	"	"
G7	7A,7B	"	"	12	"	13	"	25	"	"	"	"
G8	8A,8B	"	"	14	"	33	"	26	"	"	"	"
G9	9A,9B	"	"	34	"	35	"	37	"	"	"	"
G10	10A,10B	"	"	15	"	16	"	36	"	"	"	"
G11	11A,11B	"	"	19	"	20	"	39	"	"	"	"
G12	12A,12B	"	"	18	"	17	"	38	"	"	"	"
G13	1A,1B	"	B,D	A	4	A	5	22	"	"	"	"
G14	2A,2B	"	"	"	2	"	3	21	"	"	"	"
G15	3A,3B	"	"	"	6	"	7	24	"	"	"	"
G16	4A,4B	"	"	"	27	"	28	29	"	"	"	"
G17	5A,5B	"	"	"	32	"	31	30	"	"	"	"
G18	6A,6B	"	"	"	8	"	9	23	"	"	"	"
G19	7A,7B	"	"	"	12	"	13	25	"	"	"	"

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TABLE IX. Common mode rejection (high-level) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
G20	8A,8B	Common mode rejection, high level	B,D	A	14	A	33	26	2.4	5.0	V	9
G21	9A,9B	"	"	"	34	"	35	37	"	"	"	"
G22	10A,10B	"	"	"	15	"	16	36	"	"	"	"
G23	11A,11B	"	"	"	19	"	20	39	"	"	"	"
G24	12A,12B	"	"	"	18	"	17	38	"	"	"	"

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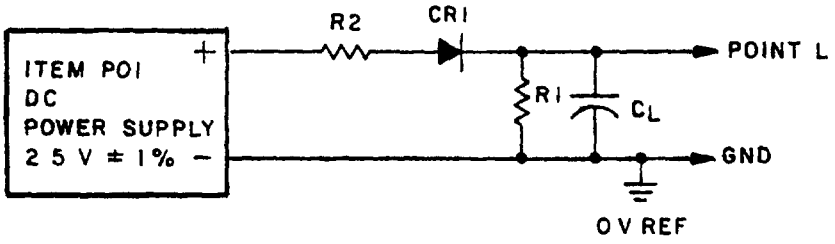


NOTE: Set item PO1-3.

Test condition	V_{IN} ±1%
E1-E12	1.9
E13-E24	1.1
F1-F12	5.6
F13-F24	9.4
G1-G12	6.4
G13-G24	8.6

* FIGURE 9. Output voltage (low-level and high-level) and common mode rejection (low-level and high-level) test circuit for tables VII, VIII, and IX.

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NOTES:
1. CR1 = 1N4150 or equivalent.
2. R1 = 1.5 kΩ ±1%, 1/8 W.
3. R2 = Select a value for a current flow of 1.6 mA ±1% from point L when L is held at 0.4 V.
4. C_L = 50 ±3 pF (including probe and parasitic capacitance) for test conditions L1-L12, M1-M12, N1-N12, and P1-P12.
5. C_L is not required for test conditions E1-E24, F1-F24, and G1-G24.
6. The maximum lead length from point L to the appropriate module pin shall be 3 inches (76.2 mm).
7. During output switching, item PO1 shall have a maximum overshoot of 500 mV.

FIGURE 10. DLC logic load circuit.

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TABLE X. Input current (low-level) test.

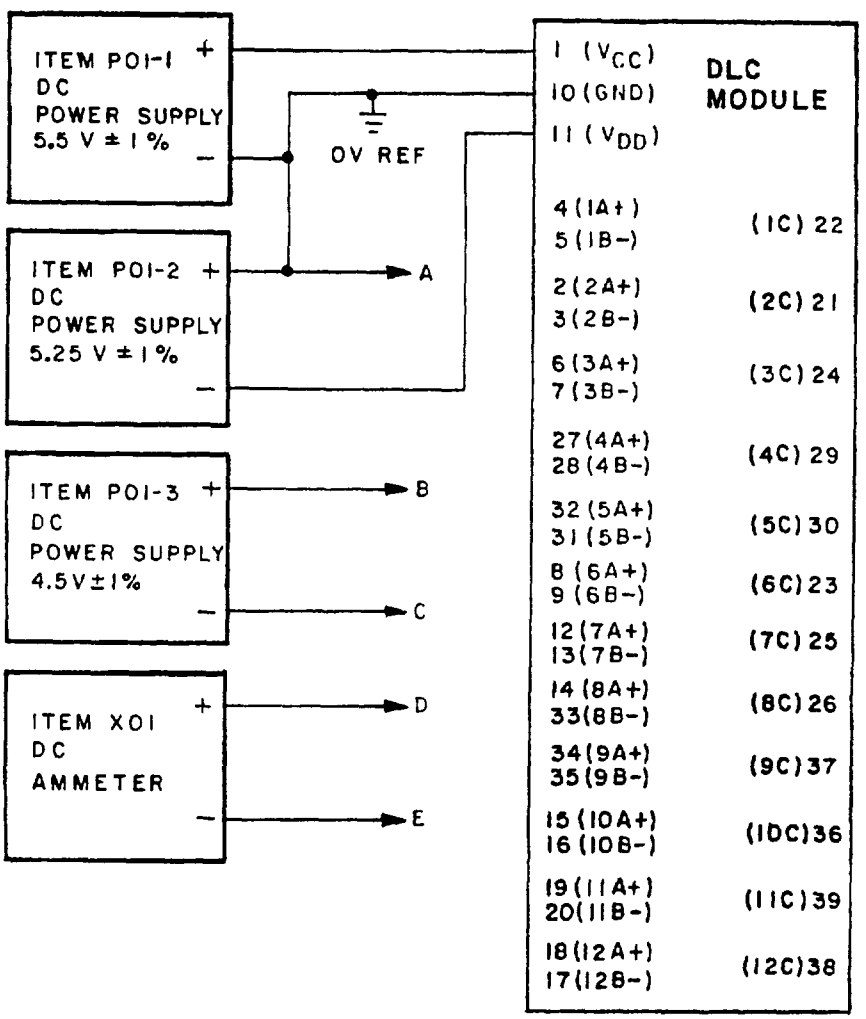
Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
H1	1A	Input current, low level	5,B	A	E	4	C	-100	-500	µA	11
H2	2A	"	3,B	"	"	2	"	"	"	"	"
H3	3A	"	7,B	"	"	6	"	"	"	"	"
H4	4A	"	28,B	"	"	27	"	"	"	"	"
H5	5A	"	31,B	"	"	32	"	"	"	"	"
H6	6A	"	9,B	"	"	8	"	"	"	"	"
H7	7A	"	13,B	"	"	12	"	"	"	"	"
H8	8A	"	33,B	"	"	14	"	"	"	"	"
H9	9A	"	35,B	"	"	34	"	"	"	"	"
H10	10A	"	16,B	"	"	15	"	"	"	"	"
H11	11A	"	20,B	"	"	19	"	"	"	"	"
H12	12A	"	17,B	"	"	18	"	"	"	"	"

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TABLE XI. Input current (high-level) test.

Test condition		Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
J1	1A	Input current, high level	5,D	Open	Open	A	4	0.1	1.5	mA	11
J2	2A	"	3,D	"	"	"	2	"	"	"	"
J3	3A	"	7,D	"	"	"	6	"	"	"	"
J4	4A	"	28,D	"	"	"	27	"	"	"	"
J5	5A	"	31,D	"	"	"	32	"	"	"	"
J6	6A	"	9,D	"	"	"	8	"	"	"	"
J7	7A	"	13,D	"	"	"	12	"	"	"	"
J8	8A	"	33,D	"	"	"	14	"	"	"	"
J9	9A	"	35,D	"	"	"	34	"	"	"	"
J10	10A	"	16,D	"	"	"	15	"	"	"	"
J11	11A	"	20,D	"	"	"	19	"	"	"	"
J12	12A	"	17,D	"	"	"	18	"	"	"	"

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CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

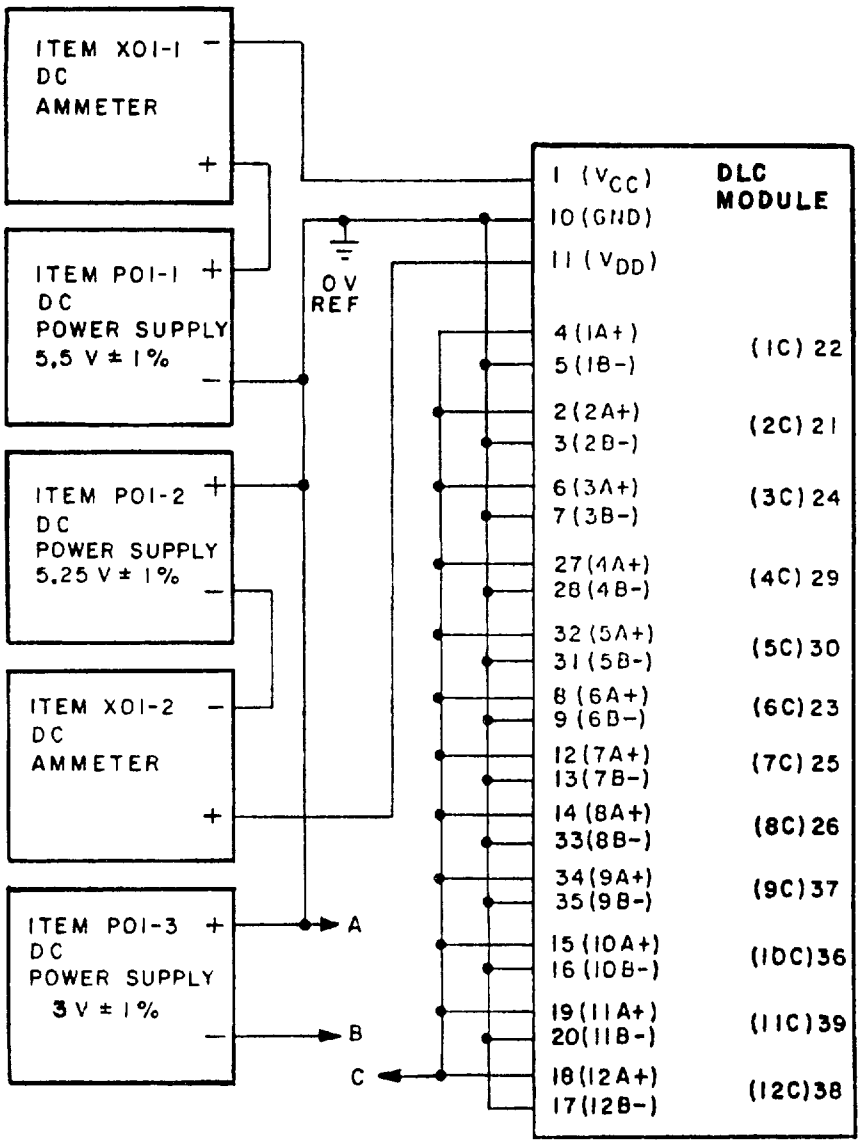
FIGURE 11. Input current (low-level and high-level) test circuit for tables X and XI.

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* TABLE XII. Power supply current (low-level and high-level) (V_{CC} and V_{DD}) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	Power supply current, low level, V_{CC}	Open	C	50	125	mA	12
K2	Power supply current, low level, V_{DD}	Open	C	70	165	"	"
K3	Power supply current, high level, V_{CC}	C	Open	50	125	"	"
K4	Power supply current, high level, V_{DD}	C	Open	70	165	"	"

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CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

FIGURE 12. Power supply current (low-level and high-level) (V_{CC} and V_{DD}) test circuit for table XII.

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TABLE XIII. Transition time (t_{TLH}) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L1	1C	Transition time (t_{TLH})	4	22	10	125	ns	13,14
L2	2C	"	2	21	"	"	"	"
L3	3C	"	6	24	"	"	"	"
L4	4C	"	27	29	"	"	"	"
L5	5C	"	32	30	"	"	"	"
L6	6C	"	8	23	"	"	"	"
L7	7C	"	12	25	"	"	"	"
L8	8C	"	14	26	"	"	"	"
L9	9C	"	34	37	"	"	"	"
L10	10C	"	15	36	"	"	"	"
L11	11C	"	19	39	"	"	"	"
L12	12C	"	18	38	"	"	"	"

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TABLE XIV. Transition time (t_{THL}) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	1C Transition time (t_{THL})	4	22	10	125	ns	13,14
M2	2C "	2	21	"	"	"	"
M3	3C "	6	24	"	"	"	"
M4	4C "	27	29	"	"	"	"
M5	5C "	32	30	"	"	"	"
M6	6C "	8	23	"	"	"	"
M7	7C "	12	25	"	"	"	"
M8	8C "	14	26	"	"	"	"
M9	9C "	34	37	"	"	"	"
M10	10C "	15	36	"	"	"	"
M11	11C "	19	39	"	"	"	"
M12	12C "	18	38	"	"	"	"

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TABLE XV. Propagation delay time (t_{PLH}) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
N1	1C	Propagation delay time (t_{PLH})	4	22	10	200	ns	13,14
N2	2C	"	2	21	"	"	"	"
N3	3C	"	6	24	"	"	"	"
N4	4C	"	27	29	"	"	"	"
N5	5C	"	32	30	"	"	"	"
N6	6C	"	8	23	"	"	"	"
N7	7C	"	12	25	"	"	"	"
N8	8C	"	14	26	"	"	"	"
N9	9C	"	34	37	"	"	"	"
N10	10C	"	15	36	"	"	"	"
N11	11C	"	19	39	"	"	"	"
N12	12C	"	18	38	"	"	"	"

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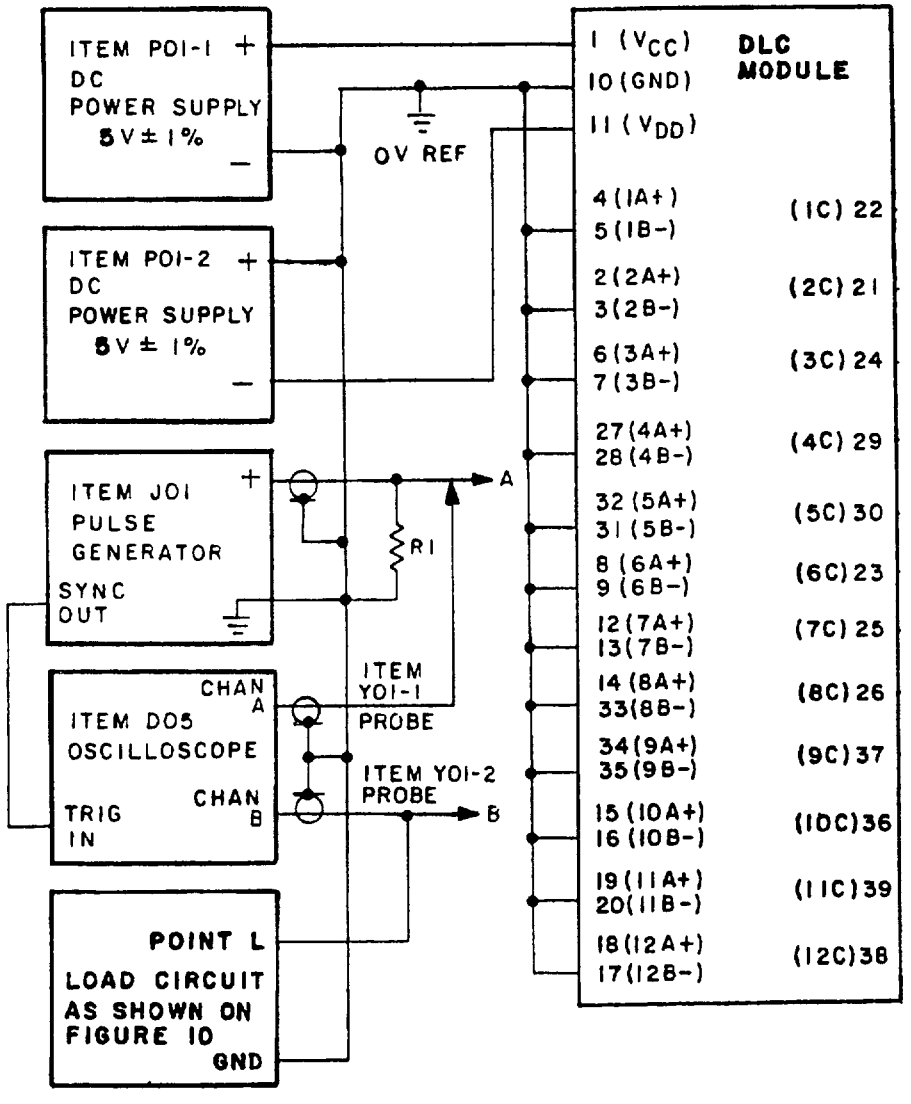
TABLE XVI. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	1C Propagation delay time (t_{PHL})	4	22	10	200	ns	13,14
P2	2C "	2	21	"	"	"	"
P3	3C "	6	24	"	"	"	"
P4	4C "	27	29	"	"	"	"
P5	5C "	32	30	"	"	"	"
P6	6C "	8	23	"	"	"	"
P7	7C "	12	25	"	"	"	"
P8	8C "	14	26	"	"	"	"
P9	9C "	34	37	"	"	"	"
P10	10C "	15	36	"	"	"	"
P11	11C "	19	39	"	"	"	"
P12	12C "	18	38	"	"	"	"

TABLE XVII. Power supply transient amplitude test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		B to pin	Min	Max		
Q1	Power supply transient amplitude, V_{CC}	1	0	400	mV	15,16
Q2	Power supply transient amplitude, V_{DD}	11	0	400	mV	15,16

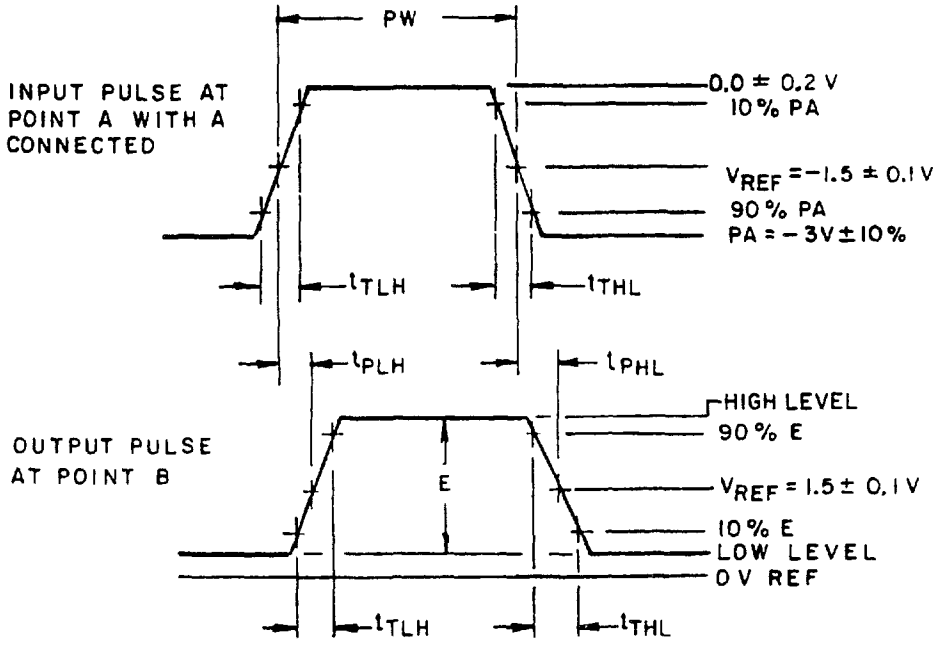
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NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XIII, XIV, XV and XVI.

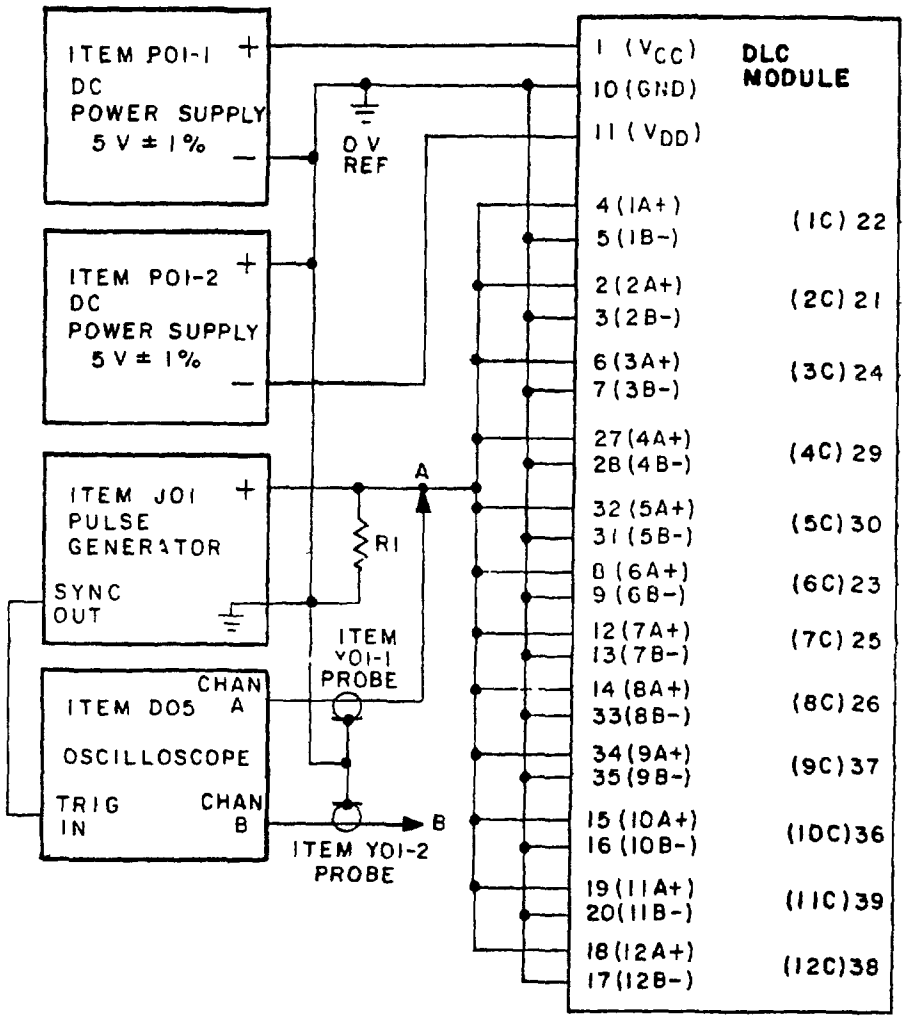
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NOTES:
1. Input pulse parameters:
   PW = 1 µs ±10%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PRR = 500 kHz ±10%.
2. Before making measurements, the output levels shall be verified to be 0.4 V maximum for the low level and 2.4 V minimum for the high level.

* FIGURE 14. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XIII, XIV, XV and XVI.

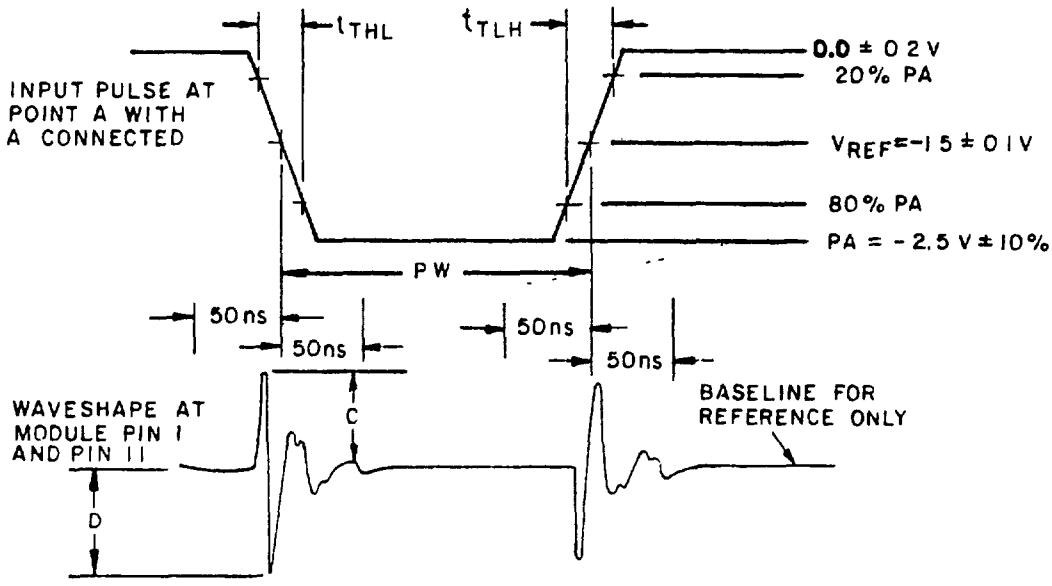
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NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 15. Power supply transient amplitude test circuit for table XVII.

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NOTES:
1. Input pulse parameters:
   PW = 1 µs ±10%.
   $t_{TLH} = t_{THL}$ = 8 ±1 ns.
   PRR = 500 kHz ±10%.
2. Observe the power supply transient at module pin 1 on item D05; the typical waveform is shown above.
3. Amplitude D shall comply with test condition Q1.
4. Amplitude D is measured from the baseline to the most negative excursion of the waveform, which may occur in time coincidence ±50 ns with the V_{REF} = -1.5 ±0.1 V point of either the leading edge or the trailing edge of the input waveform.
5. Amplitude C shall comply with test condition Q2.
6. Amplitude C is measured from the baseline to the most positive excursion of the waveform, which may occur in time coincidence ±50 ns with the V_{REF} = -1.5 ±0.1 V point of either the leading edge or the trailing edge of the input waveform.
7. Overshoot and backswing shall be 0.3 V maximum.

FIGURE 16. Power supply transient amplitude test circuit waveforms for table XVII.

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TABLE XVIII. Output crosstalk (low-level and high-level) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1A,1B	Output crosstalk, low level	2,6,27,32, 8,12,14,34, 15,19,18	Open	4	22	0	400	mV	17,18
R2	2A,2B	"	4,6,27,32, 8,12,14,34, 15,19,18	"	2	21	"	"	"	"
R3	3A,3B	"	4,2,27,32, 8,12,14,34, 15,19,18	"	6	24	"	"	"	"
R4	4A,4B	"	4,2,6,32, 8,12,14,34, 15,19,18	"	27	29	"	"	"	"
R5	5A,5B	"	4,2,6,27, 8,12,14,34, 15,19,18	"	32	30	"	"	"	"
R6	6A,6B	"	4,2,6,27, 32,12,14, 34,15,19,18	"	8	23	"	"	"	"
R7	7A,7B	"	4,2,6,27, 32,8,14,34, 15,19,18	"	12	25	"	"	"	"
R8	8A,8B	"	4,2,6,27, 32,8,12,34, 15,19,18	"	14	26	"	"	"	"
R9	9A,9B	"	4,2,6,27, 32,8,12,14, 15,19,18	"	34	37	"	"	"	"
R10	10A,10B	"	4,2,6,27, 32,8,12,14, 34,19,18	"	15	36	"	"	"	"

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TABLE XVIII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R11	11A,11B	Output crosstalk, low level	4,2,6,27, 32,8,12,14, 34,15,18	Open	19	39	0	400	mV	17,18
R12	12A,12B	Output crosstalk, low level	4,2,6,27, 32,8,12,14, 34,15,19	Open	18	38	"	"	"	"
S1	1A,1B	Output crosstalk, high level	2,6,27,32, 8,12,14,34, 15,19,18	4	Open	22	"	"	"	"
S2	2A,2B	"	4,6,27,32, 8,12,14,34, 15,19,18	2	"	21	"	"	"	"
S3	3A,3B	"	4,2,27,32, 8,12,14,34, 15,19,18	6	"	24	"	"	"	"
S4	4A,4B	"	4,2,6,32, 8,12,14,34, 15,19,18	27	"	29	"	"	"	"
S5	5A,5B	"	4,2,6,27, 8,12,14,34, 15,19,18	32	"	30	"	"	"	"
S6	6A,6B	"	4,2,6,27, 32,12,14, 34,15,19,18	8	"	23	"	"	"	"
S7	7A,7B	"	4,2,6,27, 32,8,14,34, 15,19,18	12	"	25	"	"	"	"
S8	8A,8B	"	4,2,6,27, 32,8,12,34, 15,19,18	14	"	26	"	"	"	"

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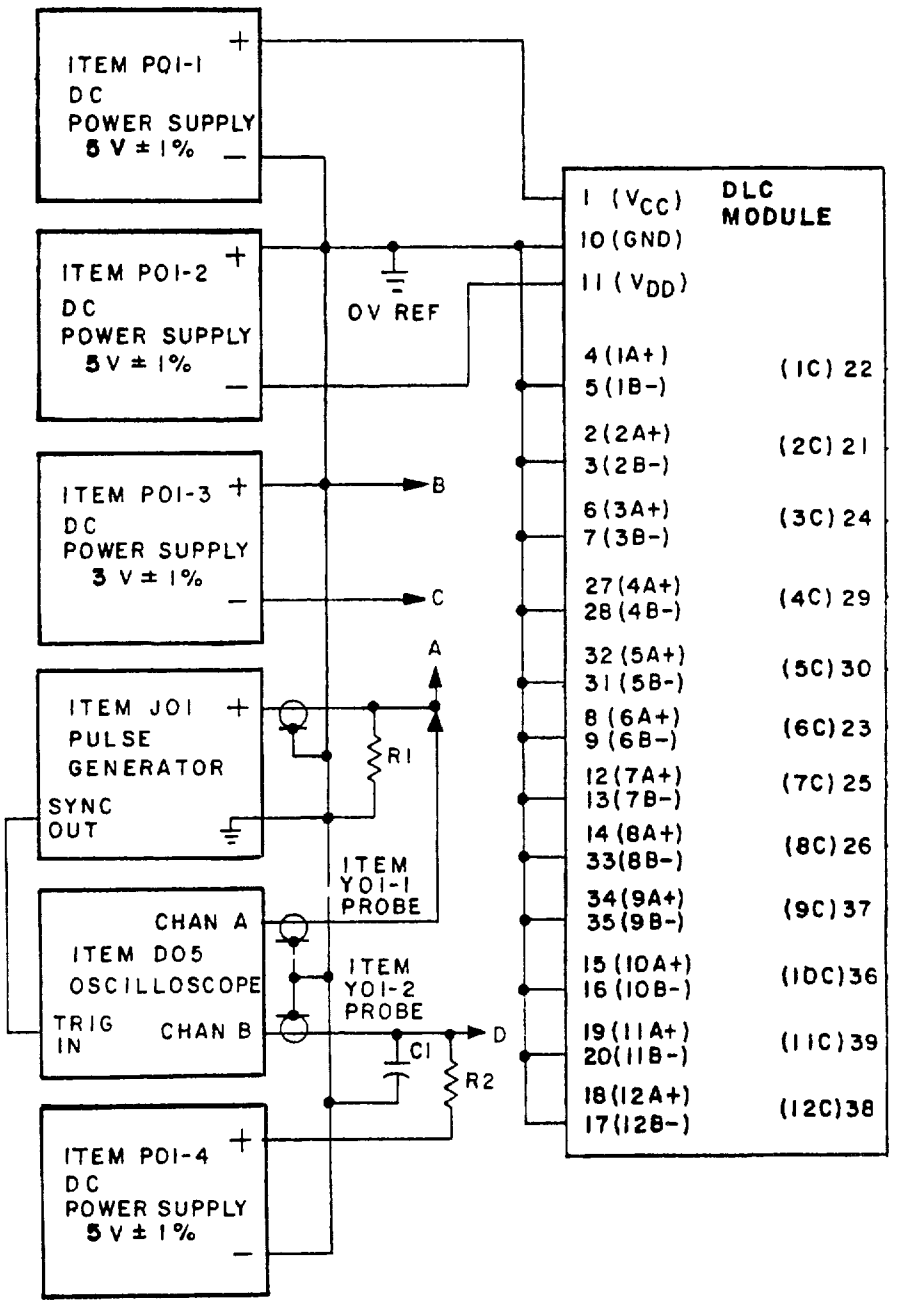
TABLE XVIII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S9	9A,9B	Output crosstalk, high level	4,2,6,27, 32,8,12,14, 15,19,18	34	Open	37	0	400	mV	17,18
S10	10A,10B	"	4,2,6,27, 32,8,12,14, 34,19,18	15	"	36	"	"	"	"
S11	11A,11B	"	4,2,6,27, 32,8,12,14, 34,15,18	19	"	39	"	"	"	"
S12	12A,12B	"	4,2,6,27, 32,8,12,14, 34,15,19	18	"	38	"	"	"	"

TABLE XIX. Power supply transient rise time test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T1	Power supply transient rise time, V_{CC}	1	10,11	0.5	1.5	ms	19,20
T2	Power supply transient rise time, V_{DD}	1,10	11	0.5	1.5	ms	19,20

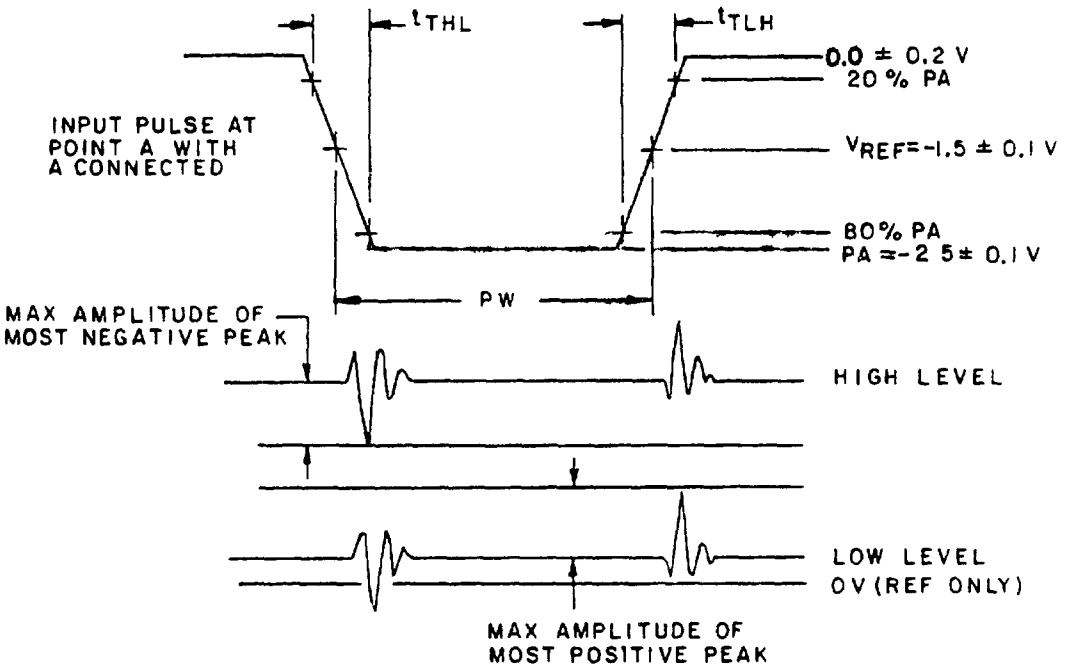
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 3.16 kΩ ±1%, 1/4 W.
3. C1 = 50 pF ± 5% (includes measured capacitance and probe capacitance).

FIGURE 17. Output crosstalk (low-level and high-level) test circuit for table XVIII.

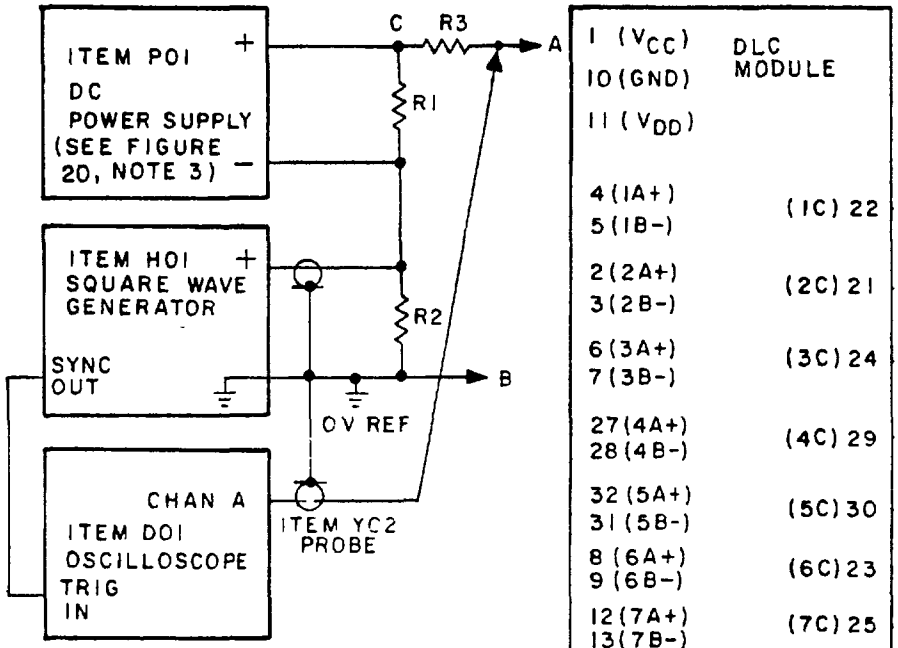
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NOTES:
1. Input pulse parameters:
   PW = 1.0 µs ±10%.
   $t_{TLH} = t_{THL}$ = 8 ±2 ns.
   PRR = 500 kHz ±10%.
2. Overshoot and backswing shall be 0.3 V maximum.

FIGURE 18. Output crosstalk (low-level and high-level) test circuit waveforms for table XVIII.

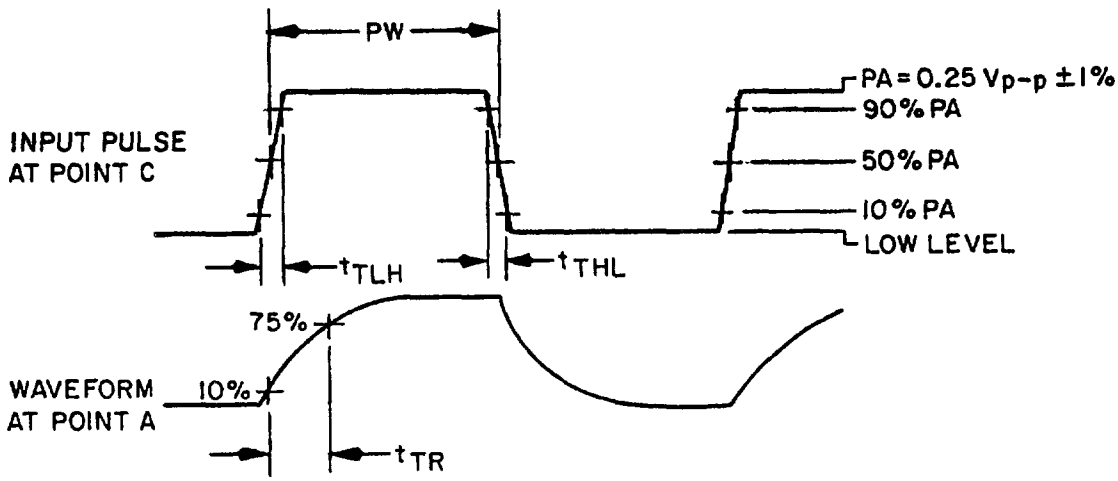
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. R3 = 511Ω ±1%, 1/4 W.

FIGURE 19. Power supply transient rise time test circuit for table XIX.

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NOTES:
1. Input pulse parameters:
   PW = 10 ms ±10%.
   $t_{TLH} = t_{THL} \leq 100$ ns (reference).
   PRR = 50 Hz (reference).
2. With point A open, adjust input as shown at point C.
3. With point A open, adjust item P01 such that the low level of the input is at 0.00 ±0.01 V.
4. Connect point A and B to module pins specified in table XIX; recheck low level and pulse parameters at point C.
5. Measure t_{TR} at point A; the value measured shall comply with test conditions T1 and T2.

FIGURE 20. Power supply transient rise time test circuit waveforms for table XIX.

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